In [2]:

```
import numpy as np
import pandas as pd
from google.colab import files
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
import pickle

uploaded = files.upload()
filename = next(iter(uploaded))

df = pd.read_excel(filename)
print(df)

df.info()

df.dropna(inplace=True)

df.info()

print(df.describe())

features = df.iloc[:, [0]].values
label = df.iloc[:, [1]].values

x_train, x_test, y_train, y_test = train_test_split(features, label,
test_size=0.2, random_state=23)

model = LinearRegression()
model.fit(x_train, y_train)

print("Training Score:", model.score(x_train, y_train))
print("Testing Score:", model.score(x_test, y_test))

print("Coefficient:", model.coef_)
print("Intercept:", model.intercept_)

pickle.dump(model, open('SalaryPred.model', 'wb'))

model = pickle.load(open('SalaryPred.model', 'rb'))

yr_of_exp = float(input("Enter Years of Experience: "))
yr_of_exp_NP = np.array([[yr_of_exp]])
Salary = model.predict(yr_of_exp_NP)

print("Estimated Salary for {} years of experience is {}:
".format(yr_of_exp, Salary))
```

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```
Saving Salary_data.xlsx to Salary_data.xlsx
    YearsExperience  Salary
0               1.1   39343
1               1.3   46205
2               1.5   37731
3               2.0   43525
4               2.2   39891
5               2.9   56642
6               3.0   60150
7               3.2   54445
8               3.2   64445
9               3.7   57189
10              3.9   63218
11              4.0   55794
12              4.0   56957
13              4.1   57081
14              4.5   61111
15              4.9   67938
16              5.1   66029
17              5.3   83088
18              5.9   81363
19              6.0   93940
20              6.8   91738
21              7.1   98273
22              7.9  101302
23              8.2  113812
24              8.7  109431
25              9.0  105582
26              9.5  116969
27              9.6  112635
28             10.3  122391
29             10.5  121872
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 30 entries, 0 to 29
Data columns (total 2 columns):
 #   Column           Non-Null Count  Dtype  
---  ------           --------------  -----  
 0   YearsExperience  30 non-null     float64
 1   Salary           30 non-null     int64  
dtypes: float64(1), int64(1)
memory usage: 612.0 bytes
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 30 entries, 0 to 29
Data columns (total 2 columns):
 #   Column           Non-Null Count  Dtype  
---  ------           --------------  -----  
 0   YearsExperience  30 non-null     float64
 1   Salary           30 non-null     int64  
dtypes: float64(1), int64(1)
memory usage: 612.0 bytes
       YearsExperience         Salary
count        30.000000      30.000000
mean          5.313333   76003.000000
std           2.837888   27414.429785
```

```
min           1.100000   37731.000000
25%           3.200000   56720.750000
50%           4.700000   65237.000000
75%           7.700000  100544.750000
max          10.500000  122391.000000
Training Score: 0.9603182547438908
Testing Score: 0.9184170849214232
Coefficient: [[9281.30847068]]
Intercept: [27166.73682891]
Enter Years of Experience: 44
Estimated Salary for 44.0 years of experience is [[435544.30953887]]:
```